RAHUL SHARMA

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar
Bachelor of Technology (B.Tech) in Computer Science & Engineering
Expected Graduation: May 2026
Cumulative Grade Point Average (CGPA): 8.2 / 10.0 (First Class with Distinction)
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Machine Learning, Deep Neural Networks, Da

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript, Bash
Frameworks & Libraries: PyTorch, FastAPI, LangGraph, LangChain, HuggingFace Transformers, OpenCV, Docker, PostgreSQL, Redis, Git
Specializations: Autonomous Multi-Agent AI Systems, Retrieval-Augmented Generation (RAG), Edge Computing, Computer Vision

ACADEMIC & OPEN SOURCE PROJECTS

1. Autonomous Agentic Research Engine (github.com/rahul-cse/agentic-research)
   - Architected a multi-agent hierarchical research framework using LangGraph and ChromaDB vector search.
   - Implemented dynamic query decomposition, automated web extraction, and fact-checking synthesis.
   - Achieved 40% reduction in research latency with parallel task routing; received 14 GitHub stars.

2. Edge-Optimized YOLO Object Detection Pipeline (github.com/rahul-cse/edge-yolo)
   - Developed an INT8-quantized object detector deployed on NVIDIA Jetson Nano running at 32 FPS.
   - Built custom TensorRT inference engine with C++ bindings for low-latency robotics perception.

3. Distributed Vector Search Microservice
   - Engineered an async vector embedding service using FastAPI and pgvector capable of handling 500 QPS.
   - Integrated semantic caching with Redis reducing LLM call costs by 65%.

HONORS & ACHIEVEMENTS

- Finalist, Smart India Hackathon (National Level Software Edition, Top 30 teams out of 2,500+)
- Recipient of NIT Jalandhar Academic Merit Certificate (Top 10% of CSE cohort)
- Scored 98.4 percentile in Joint Entrance Examination (JEE Mains)

STANDARDIZED TESTS & LANGUAGES

- IELTS Academic: Overall 7.5 (Listening: 8.5, Reading: 8.0, Writing: 7.0, Speaking: 7.0)
- Languages: English (Professional Working Proficiency), Hindi (Native), Punjabi (Native)